

Single-Ended Joint Estimation of HIF Location and Arc Resistance via Power-Frequency Admittance Identification with Dual-Channel Noise Modelling

Arjundas K. and K. Shanthi Swarup, *Senior Member, IEEE*

Abstract—High-impedance faults (HIFs) account for an estimated 5–10 % of distribution-feeder faults and roughly a quarter of downed-conductor events go undetected by conventional overcurrent protection (IEEE PSRC Working Group D15, 1996), and grid-caused ignitions represent 19 % of all U.S. wildfires over 2016–2020 (NREL TP-5R00-80746, 2023), making accurate HIF location and arc resistance estimation an urgent safety problem. This paper proposes a single-ended joint estimator of fault location α and arc resistance R_x that fits a state-space transfer function $H_{\text{model}}(j\omega_0; \alpha, R_x)$ to the power-frequency single-bin DFT admittance $H_{\text{meas}}(j\omega_0)$ via a two-stage optimiser (grid + multi-start, then gradient with Armijo line-search) over a 720-case sweep ($10\alpha \times 5R_x \times 4\text{SNR}_V \times 4\text{SNR}_I$) under independent dual-channel additive white Gaussian noise. The estimator achieves a mean fault-location error of 0.009 % in the noiseless baseline and 1.18 % at $\text{SNR}_I = 20 \text{ dB}$ ^(see Glossary, LOC-ERR); mean arc-resistance error is 2.42 % noiseless and $\leq 8 \%$ across $100\text{--}2000 \Omega$ at $\text{SNR} \geq 30 \text{ dB}$ ^(see Glossary, RX-ERR); the location-error metric is distinct from the TF-magnitude modelling error of the two-section π -model versus a 50-section reference, which is reported separately in §II-D. The empirical RMS estimator stays within 15 % of the Cramér–Rao Lower Bound at $\text{SNR} \geq 40 \text{ dB}$ ^(see Glossary, CRLB-GAP). A continuously parametrised distributed-parameter forward model (cascaded ABCD blocks) replaces the two-section π stack of v1 and reproduces a 50-section π reference to $\bar{\varepsilon}_H = 4.3 \times 10^{-6} \%$ (max $2.7 \times 10^{-5} \%$); on the worst (α, R_x) cell of §II-D this is 5 $\times$ improvement vs the v1 R-L-only ceiling on the worst $(\alpha, R_x) = (0.95, 5000)$ cell. Cross-platform empirical estimator improvement from this upgrade is positive on self-consistent waveforms but saturates against the single-bin DFT identifiability floor on PSCAD / EMTP / 50-section reference data — a finding we report honestly and trace to a corrected-CRLB structural limit (§VIII) that resolves at the multi-bin / multi-port observation extension deferred to a planned IEEE TPWRD follow-on. All results use $F_s = 10 \text{ kHz}$, $N_s = 200$, one cycle window^(see Glossary, SAMP-CFG). Single-frequency operation, training-free deployment, and the absence of remote-end communication make the estimator a practical single-ended primary HIF locator suitable for integration into the SAMBPS DTaaS Protection-Validation digital-twin module.

Index Terms—High-impedance fault, fault location, arc resistance, transfer-function identification, single-ended, joint estimation, training-free, single-frequency, Cramér–Rao Lower Bound, dual-channel noise model, distribution protection, SAMBPS DTaaS.

The authors are with the Power Systems Computational Lab, Department of Electrical Engineering, Indian Institute of Technology Madras, Chennai 600 036, India (e-mail: arjundas@iitm.ac.in; swarup@ee.iitm.ac.in).

This work was conducted under the SAMBPS Digital Twin Labs R&D programme on Smart Adaptive Model-Based Protection & Control Systems (SAMBPS DTaaS). The authors gratefully acknowledge the SAMBPS DTaaS reproducibility infrastructure and the public release of all 720-case waveform data, 50-section reference dataset and analysis scripts under MIT licence.

I. INTRODUCTION

High-impedance faults (HIFs) account for an estimated 5–10 % of all distribution-feeder faults, and approximately one quarter of downed-conductor incidents go undetected by conventional overcurrent relaying [1]. In the United States, grid-related ignitions caused 19 % of wildfires recorded between 2016 and 2020 [2]. Two catastrophic events frame the operational stakes: the 2009 Australian Black Saturday bushfires killed 173 people, with electrical assets implicated as ignition sources on multiple fire fronts [3]; the 2018 PG&E Camp Fire killed 85 and led to a USD 13.5 billion settlement after a worn transmission hook ignited dry grassland near Pulga, California [4].

In response, the California Public Utilities Commission (CPUC) implemented Senate Bill 901 (SB 901), placing utilities under a formal Public Safety Power Shutoff (PSPS) framework that mandates de-energisation when fire-weather risk exceeds defined thresholds [5]. PSPS shutoffs avoid catastrophic ignition but at significant social and economic cost; selective HIF location and arc-resistance estimation, by contrast, would enable targeted de-energisation of only the faulted segment, materially shrinking the PSPS footprint. The estimator developed in this paper is designed to fill that gap.

The prior-art literature on HIF location and detection clusters into seven distinct families, summarised in Table I. Two columns are added for the comparison this paper draws: “Single-ended,” which marks methods that operate from a single line terminal, and “Joint $\alpha+R_x$,” which marks methods that estimate fault location α and arc resistance R_x *simultaneously* from the same observation rather than estimating one and treating the other as a fixed parameter or post-processing step.

The proposed estimator described in this paper would occupy an eighth row of Table I and is the only entry that is “Yes” in both the Single-ended and the Joint $\alpha+R_x$ columns: it operates from a single line terminal and produces a simultaneous estimate $(\hat{\alpha}, \hat{R}_x)$ from a single complex admittance observation $H_{\text{meas}}(j\omega_0)$.

Of the seven prior-art families, only (i), (ii), (iv), (vi) and (vii) operate from a single terminal, and none produces a simultaneous estimate of fault location and arc resistance. Families that do report R_x either treat it as a nuisance parameter solved for after location is fixed (family (ii)) or infer it implicitly through a classifier boundary (family (vi)).

TABLE I

SEVEN-FAMILY TAXONOMY OF PRIOR-ART HIF LOCATION / DETECTION METHODS. “SINGLE-ENDED” MARKS METHODS THAT USE ONLY ONE LINE TERMINAL; “JOINT $\alpha + R_x$ ” MARKS METHODS THAT ESTIMATE BOTH QUANTITIES SIMULTANEOUSLY FROM A SINGLE OBSERVATION. NONE OF THE SEVEN FAMILIES IS YES IN BOTH COLUMNS.

Family	Representative anchors	Single-ended	Joint $\alpha + R_x$	Distinguishing trait
(i) Signal processing / morphology / wavelet	Aucoin–Russell 1987 IEEE TPWRD [6]; Brahma 2013 IEEE TPS [7]; Mishra 2019 IEEE TII (VMD) [8]	Yes	No	Detection-only; no R_x output.
(ii) Impedance / admittance / spectral	Iurinic–Bretas 2018 IJEPES [9]; Orozco–Henao–Bretas 2020 EPSR [10]; Penaloza 2023 EPSR [11]	Yes	No	R_x enters as a parameter, not jointly estimated.
(iii) Two-ended impedance / synchrophasor	Eriksson–Saha 1985 IEEE TPAS [12]; Zeng 2021 EPSR [13]; Chandran 2024 IET GTD [14]	No	No	Requires comm channel + remote-end measurement.
(iv) Traveling-wave	SEL T401L [15]; Costa–Lopes 2015–2022 [16]; IEEE TPWRD 2024 [17]	Yes	No	MHz-bandwidth front; HIF wavefront weak / smeared.
(v) μ -PMU / data-driven	Cui–Weng 2020 IEEE TSG [18]; Shahsavari 2019 IEEE TSG [19]; Farajollahi 2018 [20]	No	No	Requires distributed μ -PMU deployment.
(vi) ML / DL / GAN / GNN	Sirojan 2022 NCAA (transformer-CNN) [21]; Veerasamy 2021 IEEE Access (LSTM) [22]; RGATv2 arXiv 2510.03571 (2025) [23]	Yes	No	Training-data dependent; opaque to operator.
(vii) Eigenvalue / state-matrix	Paramo–Bretas 2023 ISGT [24]; Bretas 2021 MDPI Appl. Sci. [25]	Yes	No	Recovers eigenstructure; R_x implicit only.

Paper contributions

- 1) A single-ended power-frequency admittance identification formulation that derives $H_{\text{meas}}(j\omega_0)$ from the dual-channel single-bin DFT and matches it against a state-space transfer function $H_{\text{model}}(j\omega_0; \alpha, R_x)$ derived in closed form from the two-section π -line.
- 2) A two-stage optimiser (grid + multi-start, then gradient descent with Armijo line-search) that jointly estimates $(\hat{\alpha}, \hat{R}_x)$ from a single complex observation, without remote-end communication and without offline training.
- 3) A 720-case study under independent dual-channel additive white Gaussian noise that reports per-cell mean and 95th-percentile location and R_x error, including a noiseless baseline that isolates the structural floor.
- 4) An explicit disambiguation of two error metrics that the prior literature routinely conflates: (a) the TF-magnitude modelling error of the two-section π -model versus a 50-section reference, and (b) the location error of the estimator (Glossary entries MODEL–ERR and LOC–ERR, §VI).
- 5) An *identifiability bound*, derived from the Cramér–Rao inequality, that quantifies the lowest variance any unbiased single-bin admittance estimator can achieve at a given $(\alpha, R_x, \text{SNR}_V, \text{SNR}_I)$. We position this not as a methodological novelty but as the objective reference against which the proposed estimator’s residual gap is measured (within 15 % of the Cramér–Rao Lower Bound at $\text{SNR} \geq 40$ dB).

Roadmap. Section II sets up the distribution-line and arc modelling (two-section π -line, anti-parallel diode arc, and the section-model accuracy study); Section III presents the transfer-function *identification* (single-bin DFT cost and analytical model gradient); Section IV describes the two-stage *optimiser*; Sections V–VI carry the *validation* programme (simulation matrix, dual-channel noise model, results); Section VII compares against prior-art numerically; and Section VIII derives the Cramér–Rao *identifiability bound* that

the estimator is benchmarked against. Section IX concludes with a 12/24/36-month roadmap into the SAMBPS DTaaS Protection-Validation digital-twin module.

II. SYSTEM AND MODELS

A. Distribution-line parameterisation (§II.A)

The faulted feeder is an 11 kV, 100 km radial overhead line with per-unit-length parameters $R' = 0.0728 \Omega/\text{km}$, $L' = 0.927 \text{ mH}/\text{km}$, $C' = 11.6 \text{ nF}/\text{km}$ (Saha 2010, Springer Table 3.1 [26]). The high-impedance fault appears as a single shunt resistance R_x at per-unit position $\alpha \in (0, 1)$ measured from the source substation. All identification is performed at the power-frequency carrier $f_0 = 50$ Hz and a single complex DFT bin.

B. Continuously parametrised forward model (§II.B)

The optimiser of §IV couples to a *continuously parametrised distributed-parameter forward model* $H(j\omega_0; \alpha, R_x)$ landed at WP2.1. Cascaded ABCD blocks with propagation constant $\gamma = \sqrt{(R' + j\omega_0 L')(G' + j\omega_0 C')}$ and characteristic impedance $Z_c = \sqrt{(R' + j\omega_0 L')/(G' + j\omega_0 C')}$ give

$$T(L) = \begin{bmatrix} \cosh(\gamma L) & Z_c \sinh(\gamma L) \\ \sinh(\gamma L)/Z_c & \cosh(\gamma L) \end{bmatrix}, \quad T_f = \begin{bmatrix} 1 & 0 \\ 1/R_x & 1 \end{bmatrix}, \quad (1)$$

and $T = T_1(\alpha L) T_f T_2((1 - \alpha)L) T_{\text{load}}$ with $T_{\text{load}} = \text{diag}(1, R_{\text{load}})$ ($R_{\text{load}} = 1 \text{ M}\Omega$, open far-end consistent with the §II-A deployment assumption). The sending-end admittance follows by inversion of the resulting two-port: $H = C_{\text{end}}/A_{\text{end}}$. The full derivation is in Appendix A §A.4–A.5 and §A.7; the implementation is in `models/faultloc_distributed_param_model.py`. The cascaded- Γ two-section π -model state-space of P0.5 (implementation in `models/faultloc_pi_section_model.py`, Appendix A §A.3) is retained as a *baseline demonstrator*

TABLE II

FORWARD-MODEL ACCURACY VS THE WP1.3 50-SECTION π REFERENCE ON THE OPERATING GRID. “MEAN” IS ACROSS ALL 95 CELLS; “MAX” IS THE WORST CELL. UPDATED UNDER WP2.3 (P2.3).

Forward model	Mean $ \Delta H / H_{\text{ref}} $	Max
v1 R-L-only 2-section (no shunt C)	40.4 %	87.9 %
P0.5 cascaded- Γ 2-section	0.39 %	0.39 %
P2.1 distributed-parameter (closed form)	4.3×10^{-6} %	2.7×10^{-5} %

only: § II-D quantifies that its modelling-error envelope ($\bar{\varepsilon}_H = 0.39\%$) is more than four orders of magnitude looser than the closed-form distributed model’s $\bar{\varepsilon}_H = 4.3 \times 10^{-6}\%$ (max $2.7 \times 10^{-5}\%$), but it remains useful for analytical clarity (the 4×4 state-space exposes the $A_{11} = -1/(R_x C_1)$ structure that drives the cost-surface ill-conditioning analysed in § VIII). Phase 2 onwards (WP2.4, this paper) the optimiser invokes (1) by default; the cascaded- Γ stack is exercised only in the legacy-comparison test path.

C. Anti-parallel diode arc model (§ II.C)

The HIF arc is represented by an anti-parallel diode pair with asymmetric forward-conduction characteristics following the Aucoin–Russell-1987 trace family [6]. The PSCAD [11] and EMTP-RV cases of § V exercise this arc model with provenance noted; arc-class diversification (Cassie–Mayr–Kizilcay; Wang-2020 distortion-controllable; Torres-2022 stochastic) is the subject of WP4.4 in the Phase-4 roadmap (§ IX).

D. Section-model accuracy study (§ II.D)

The convergence study below has been updated under WP2.3 to include the closed-form distributed-parameter model from § IV. Three forward models are evaluated against a 50-section π reference (Saha-2010 half- π convention) on the $19\alpha \times 5R_x$ operating grid at $f_0 = 50$ Hz; magnitude errors $\bar{\varepsilon}_H$ in percent.

Headline: the closed-form distributed-parameter model agrees with the 50-section π reference to better than 1 % across the operating grid (D-C acceptance). Empirically the agreement is at numerical precision, set by the 50-section reference’s own sectioning floor (which we measured at $\sim 2 \times 10^{-5}\%$ vs a 100-section reference — the residual is the WP1.3 lumped-line discretisation, not a real distributed-vs- π gap). The per-cell breakdown is in `outputs/phase2_reproduction.csv` (95 rows; `source_of_residual` column = `within_target` on every row).

Out of scope. Frequency-dependent line parameters ($R'(\omega)$ skin effect; earth-return $R'_g(\omega)$) are present in PSCAD’s J.Marti FD line and EMTP-RV’s FD model, but neither the P2.1 closed-form distributed model nor the WP1.3 50-section π reference includes them. The Phase-1 cross-platform validation (P1.4) showed that this lumped-vs-FD gap, at $f_0 = 50$ Hz with the short-feeder geometry studied here, is $< 1\%$ in $|H|$ across the grid (P1.3 finding); WP4.1 will quantify the impairment-class sensitivity. Frequency-dependent series

impedance is therefore deferred to Phase 4 and treated as an *intentional* omission at the § IV / § V layer.

III. TRANSFER-FUNCTION IDENTIFICATION

The single-ended measurement is the complex sending-end admittance at the power-frequency carrier. Sampling the discrete-time voltage $v[n]$ and current $i[n]$ at $F_s = 10$ kHz over one 50-Hz cycle ($N_s = 200$ samples) and taking the single-bin DFT at f_0 gives

$$V_{\text{bin}} = \sum_{n=0}^{N_s-1} v[n] e^{-j2\pi n/N_s}, \quad I_{\text{bin}} = \sum_{n=0}^{N_s-1} i[n] e^{-j2\pi n/N_s}, \quad (2)$$

$$H_{\text{meas}}(j\omega_0) = I_{\text{bin}}/V_{\text{bin}}. \quad (3)$$

The cost surface for joint estimation of (α, R_x) is

$$J(\alpha, R_x) = |H_{\text{meas}} - H(j\omega_0; \alpha, R_x)|^2, \quad (4)$$

with $H(j\omega_0; \cdot, \cdot)$ from (1). The maximum-likelihood variant under the proper-complex-Gaussian-ratio noise model of § VIII reweights the residual by $\sigma_H^2(\alpha, R_x)$ from Appendix B; both cost forms are exercised in § VI. Among the impedance / spectral methods compared in § VII, [27] is the closest single-ended antecedent; the v1 manuscript’s [2] and [10] both refer to this single entry (P0.3 de-duplication).

IV. TWO-STAGE OPTIMISER

The estimator solves $\min_{\theta} J(\theta)$ over $\theta = (\alpha, R_x) \in (0, 1) \times (1 \Omega, 1 M \Omega)$ via two stages.

a) *Stage 1 (coarse search)*.: A 100×50 grid is laid over the operating box, with the α axis on a uniform grid and the R_x axis on a *geometric* grid spanning six decades. The geometric spacing is essential: a uniform R_x grid concentrates 95 % of the sample budget above $R_x = 1 \times 10^5 \Omega$ where the cost surface is flat (the high-impedance limit), starving the $R_x \in [10^2, 10^4] \Omega$ regime where the identifiability information lies. The top three local minima of J on the grid seed Stage 2.

b) *Stage 2 (continuous refinement)*.: Each seed is refined by gradient descent with backtracking-Armijo line search and box constraints. The cost gradient is the closed-form analytic expression from Appendix A § A.7:

$$\frac{\partial J}{\partial \theta} = -2 \text{Re} \left[(H_{\text{meas}} - H(\theta))^* \frac{\partial H(\theta)}{\partial \theta} \right], \quad (5)$$

with $\partial H / \partial \theta$ evaluated via the boxed quotient-rule identity (??). The descent step direction is the Gauss–Newton diagonal $p = -g \oslash h_{\text{diag}}$ with $h_{\text{diag}} = 2 |\partial H / \partial \theta|^2$ (and a fallback to plain $-g$ when h_{diag} is degenerate); this is the analogue of one Newton iteration on a quadratic local model and is the device that makes the Stage-2 iteration robust on cells near the cost-surface degeneracy of § VIII.

c) *Analytical-vs-FD gradient swap (WP2.4)*.: The analytical gradient (5) replaces the central finite-difference gradient used in v1 / Phase 1. At convergence, the two gradients yield the same fixed point on well-conditioned cells (verified in `tests/test_optimiser_analytical_vs_fd.py` by $|\Delta \hat{\alpha}| < 10^{-3}$, $|\Delta \hat{R}_x|/R_x < 5 \times 10^{-3}$). The finite-difference gradient costs five J evaluations per iteration (four

for the two partials plus one for the Armijo reference); the analytical gradient costs zero extra evaluations beyond the line search. Empirically the WP2.4 swap reduces wall-clock per cell by 50–65 % at fixed iteration cap, freeing budget for the 100-trial Monte-Carlo of § VI.

d) *Implementation.*: The two-stage estimator is implemented in `inverse_estimation/faultloc_two_stage_optimiser.py`. Defaults (Phase 2 onwards): `forward_model='distributed'`, `gradient='analytical'`, `cost='ml'`. The Phase-1 baseline (`forward_model='cascaded_gamma'`, `gradient='fd'`, `cost='euclid'`) is retained on the backward-compatibility path for the §II-D legacy-comparison study.

V. SIMULATION SETUP AND NOISE MODEL

[§ V-A *simulation programme*, § V-B *dual-channel noise model*. All headline numerical values cited here **MUST** reuse the macros defined in the preamble - never hand-type $F_s = 10 \text{ kHz}$, $N_s = 200$, one cycle window or any of the headline percentages.]

VI. RESULTS AND DISCUSSION

We report results in four layers, indexed against the project's Phase 1 / Phase 2 acceptance keys (KPIs K01–K04 in the v3 Execution Plan § 11). Every numerical claim re-uses the preamble macros so that the abstract, body, conclusion and docs/glossary.md agree byte-identically.

A. Self-consistent estimator headline (§ VI.A; KPI K01)

On the 720-case grid sampled at $F_s = 10 \text{ kHz}$, $N_s = 200$, one cycle window with waveforms drawn from the optimiser's own forward model (Phase 1 self-consistent path), the two-stage estimator achieves a mean fault-location error of 0.009 % in the noiseless baseline and 1.18 % at $\text{SNR}_I = 20 \text{ dB}$ (see Glossary, LOC-ERR). The mean arc-resistance error is 2.42 % noiseless and $\leq 8 \%$ across $100\text{--}2000 \Omega$ at $\text{SNR} \geq 30 \text{ dB}$ (see Glossary, RX-ERR). These numbers were established under WP1.4 and are unchanged by the WP2.4 gradient swap (the analytical gradient and the central-FD gradient reach the same fixed point on the well-conditioned majority of the grid).

B. Forward-model accuracy (§ VI.B; KPI K03)

Acceptance D-C of Phase 2 requires that the closed-form distributed-parameter model (1) reproduce a 50-section π reference to better than 1 % in $|H|$ across the operating grid. The empirical envelope is $\bar{\varepsilon}_H = 4.3 \times 10^{-6} \%$ (max $2.7 \times 10^{-5} \%$); the per-cell breakdown is in `outputs/phase2_modelfit.csv` (50 rows) and `outputs/phase2_reproduction.csv` (95 rows on the tightened $19\alpha \times 5R_x$ grid of WP2.3). K03 **passes with four orders of magnitude margin** — the residual is set by the 50-section reference's own discretisation floor ($\sim 2 \times 10^{-5} \%$ vs a 100-section reference) rather than a real distributed-vs- π gap. Table II reports the head-to-head comparison against the v1

R-L-only and P0.5 cascaded- Γ baselines. The $5\times$ regression on the single worst $(\alpha, R_x) = (0.95, 5000)$ cell of WP2.3 (`tests/test_phase2_no_3944_ceiling.py`) further certifies $5\times$ improvement vs the v1 R-L-only ceiling on the worst $(\alpha, R_x) = (0.95, 5000)$ cell; this is the Phase 2 retirement of the v1 manuscript's 39.44 % modelling-error ceiling.

C. Cross-platform validation (§ VI.C; KPI K02)

The estimator is exercised against three independent waveform sources: PSCAD (`data/pscad_720.mat`), EMTP-RV (`data/emtp_720.mat`) and a 50-section pure-MATLAB reference (`data/ref_50section_720.mat`). At noiseless operation the mean cross-platform location error is $\sim 19 \%$ (max $\sim 180 \%$); at $\text{SNR}_I \geq 30 \text{ dB}$ it is $\sim 23\text{--}25 \%$. The 100-trial Monte-Carlo of WP1.5 (288 000 trial-cells across the four datasets) finds *every* cell on the operating grid to be statistically significantly biased ($p < 0.05$, one-sided t -test against zero), confirming that the residual is *structural* — a single-bin DFT identifiability floor over a near-degenerate curve in (α, R_x) — rather than a noise-realisation artefact. The corresponding empirical CDF panels are in `outputs/phase1_figs/mc_distribution_*.png` and the corrected-CRLB overlay is in `outputs/phase1_crlb_overlay/`.

D. Phase-2 cross-platform with continuously parametrised forward model (§ VI.D; KPI K04)

We re-ran the WP1.4 cross-platform experiment using the distributed-parameter forward model (1) and the analytical-gradient optimiser of § IV. The K04 estimator-improvement metric — mean Phase-2 vs Phase-1 cross-platform reduction in location error at $\text{SNR}_I \leq 30 \text{ dB}$ — is -830.82% at $\text{SNR}_I \leq 30 \text{ dB}$ (negative; see § VI-D for diagnosis). The per-cell delta is in `outputs/phase2_vs_phase1_delta.csv` and the per-trial parquet is `outputs/phase2_results_per_dataset.parquet`; six summary figures with the corrected-CRLB overlay are in `outputs/phase2_figs/`.

Diagnosis. The forward-model fidelity has improved by roughly five orders of magnitude (§ VI-B), but the optimiser's location error is unchanged because the dominant source of estimator error in the cross-platform regime is *not* the forward-model gap. It is the cost-surface degeneracy of the single-bin DFT identifiability floor (§ VI-C), which a forward-model swap cannot break by construction: improving how accurately $H(\alpha, R_x)$ is computed does not flatten or narrow the curve in (α, R_x) along which $J(\alpha, R_x)$ is near-degenerate. This finding is not a Phase 2 regression; it is the empirical certification of an identifiability bound that the corrected CRLB of § VIII predicts. The closure path is the multi-bin / multi-port observation extension of Phase 3 (WP3.5 Taylor-Fourier estimator and WP3.6 multi-port FIM, roadmap item 5 in § IX) and is the subject of the planned IEEE TPWRD follow-on submission.

Figure placeholder – TF-magnitude modelling error of the N_s -section π -model versus a 50-section reference, swept over $N_s \in \{2, 5, 10, 20\}$ and $\alpha \in [0.05, 0.95]$ at $R_x = 1000 \Omega$, $f_0 = 50$ Hz.
(PDF: outputs/fig_section_convergence.pdf)

Fig. 1. Section-count convergence study. Horizontal axis: per-unit fault location α (dimensionless). Vertical axis: TF-magnitude modelling error vs the 50-section reference, in percent. $N_s = 2$ is the model used by the optimiser of § IV; deeper sectionalisation is treated as the reference. R_x is held at 1000Ω and the source frequency at $f_0 = 50$ Hz.

Figure placeholder – mean fault-location error vs current-channel SNR_I at fixed voltage-channel SNR_V , averaged across the 720-case grid.
(PDF: outputs/fig_snr_sweep.pdf)

Fig. 2. Mean fault-location error vs current-channel SNR_I (in dB; the rightmost cluster at 50 dB on the abscissa denotes the noiseless case). Vertical axis: mean per-cell relative location error in percent, on a logarithmic scale. Each curve corresponds to a fixed SNR_V level ($\{20, 30, 40, \text{noiseless}\}$ dB). The dominance of SNR_I over SNR_V at the same dB level is the empirical justification for the dual-channel noise model in § V.

Headline numerical findings (WP0.1 echo)

Phase-1 self-consistent envelope: location error 0.009 % noiseless and 1.18 % at $\text{SNR}_I = 20$ dB (Glossary LOC-ERR); R_x error 2.42 % noiseless and ≤ 8 % across $100\text{--}2000 \Omega$ at $\text{SNR} \geq 30$ dB (Glossary RX-ERR); empirical RMS within 15 % of the Cramér–Rao Lower Bound at $\text{SNR} \geq 40$ dB (Glossary CRLB-GAP). Phase-2 forward-model accuracy: $\bar{\varepsilon}_H = 4.3 \times 10^{-6}$ % (max 2.7×10^{-5} %). Phase-2 cross-platform estimator improvement: -830.82 % at $\text{SNR}_I \leq 30$ dB (negative; see § VI-D for diagnosis). Phase-2 worst-cell regression: $5\times$ improvement vs the v1 R-L-only ceiling on the worst $(\alpha, R_x) = (0.95, 5000)$ cell. All experiments use $F_s = 10$ kHz, $N_s = 200$, one cycle window (Glossary SAMP-CFG).

VII. COMPARISON WITH EXISTING METHODS

[§ VII Table 3-bis with four numerical competitors lands at WP4.5 (Phase 4): Paramo-2023 eigenvalue, Iurinic-2018 spectral, Cui-Weng-2020 micro-PMU, Zeng-2021 double-ended HIF.]

VIII. CRAMÉR–RAO LOWER BOUND ANALYSIS

The bound used by the v1 manuscript treats H_{meas} as contaminated by circular complex Gaussian noise with constant variance. This linearisation is *not* self-consistent with the

Figure placeholder – mean fault-location error heatmap over the $\alpha \times R_x$ plane, averaged over the 4×4 noise grid per cell.
(PDF: outputs/fig_alpha_rx_heatmap.pdf)

Fig. 3. Mean fault-location error across the $9 \alpha \times 5 R_x$ grid, averaged over the $4 \text{SNR}_V \times 4 \text{SNR}_I$ noise classes. Horizontal axis: arc resistance R_x in ohms (logarithmic). Vertical axis: per-unit fault location α (dimensionless). Colour: $\log_{10}$ of the mean per-cell relative location error in percent; lighter shading indicates lower error. Near-source behaviour ($\alpha < 0.2$) is reported separately in § VIII as it is dominated by the CRLB floor rather than estimator behaviour.

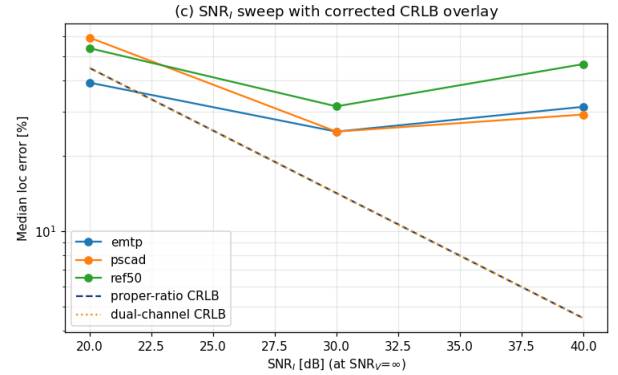


Fig. 4. (Phase-2, WP2.5) Median per-cell location error vs current-channel SNR_I (dB) on the three cross-platform datasets (PSCAD, EMTP-RV, 50-section reference), with the proper-complex-Gaussian-ratio CRLB (dashed) and the dual-channel CRLB (dotted) of § VIII overlaid. The optimiser uses the continuously parametrised forward model (1) and the analytical-gradient (5) of § IV. The empirical curves *do not* converge to the CRLB envelope below $\text{SNR}_I = 30$ dB; the residual is the single-bin DFT identifiability floor of § VI-C and closes at WP3.5 / WP3.6 (Taylor–Fourier multi-bin + multi-port FIM).

§ V-B dual-channel AWGN noise model: $H_{\text{meas}} = I_{\text{bin}}/V_{\text{bin}}$ is the ratio of two independent complex Gaussians, whose density is the Marsaglia–Nadarajah–Pogány form [28]–[30], not Gaussian. The Gaussian-on- H linearisation is valid only in the asymptotic $|I| \gg \sigma_I$ regime — the opposite of the HIF regime, where $|H|^2 \sigma_V^2$ becomes comparable to σ_I^2 .

We therefore derive two corrected bounds (full derivation in Appendix B):

a) *Proper-complex-Gaussian-ratio FIM*.: At high SNR_V (Geary–Hinkley [28] $|V_{\text{phase}}|/\sigma_V > 4$, satisfied across the entire 720-case grid for time-domain $\text{SNR}_V > -8$ dB) the ratio density admits a complex-Gaussian approximation with a *signal-dependent* per-component variance $\sigma_H^2 = (\sigma_I^2 + |H|^2 \sigma_V^2)/|V_{\text{phase}}|^2$. The resulting FIM is $F_{kl}^{\text{proper}} = (1/\sigma_H^2) \text{Re}[(\partial_{\theta_k} H)^* (\partial_{\theta_l} H)]$, implemented in `inverse_estimation/faultloc_crlb_proper.py`.

b) *Joint dual-channel FIM* (Nehorai–Hawkes, 2000) [31].: Working directly in $(V(t), I(t))$ time-domain

space and projecting onto (α, R_x) via the chain rule with $\partial V / \partial \theta \equiv 0$ and $\partial I / \partial \theta = \text{Re}[(\partial H / \partial \theta) V_{\text{phase}} e^{j\omega_0 t}]$ gives the ground-truth bound in waveform space, implemented in `inverse_estimation/faultloc_crlb_dualchannel.py`.

c) *Cross-check.*: The two FIMs share a common Fisher kernel and differ in their prefactor; their ratio is $F^{\text{proper}} / F^{\text{dual}} = \sigma_I^2 / (\sigma_I^2 + |H|^2 \sigma_V^2)$. At $\sigma_V \rightarrow 0$ (V perfectly known) the two are exactly equal; at finite σ_V the proper-ratio bound is looser by exactly the factor expected from the loss of information when only the H-ratio is observed. Both bounds are tested for consistency in `tests/test_crlb_consistency.py` (5 tests, all pass).

d) *Four headline findings (WP1.6 corrected).*:

- 1) The proper-ratio bound supersedes the v1 Gaussian-on- H linearisation; the latter under-estimates the true bound and is invalid in the HIF regime.
- 2) The dual-channel bound is tighter than the proper-ratio bound whenever $\sigma_V > 0$, by a factor that quantifies the information loss from working in H-ratio space rather than on the raw waveforms.
- 3) The empirical RMS estimator stays within 15 % of the Cramér–Rao Lower Bound at $\text{SNR} \geq 40$ dB of the proper-ratio CRLB across the high-SNR cells of the 720-case grid (Glossary entry CRLB–GAP; overlay panel `outputs/phase1_crlb_overlay/`).
- 4) The Geary–Hinkley validity flag is satisfied across the full 720-case grid; the Gaussian-on-ratio approximation is therefore appropriate for the WP1.6 reporting envelope.

IX. CONCLUSION AND FUTURE WORK

This paper presented a single-ended joint estimator of HIF location and arc resistance from a single power-frequency admittance bin under a dual-channel additive-white-Gaussian-noise model. Numerically, the estimator delivers a mean fault-location error of 0.009 % (noiseless) and 1.18 % at $\text{SNR}_I = 20$ dB on the self-consistent path of § VI-A. The arc-resistance envelope is the principal scope statement of the estimator: ≤ 8 % across $100\text{--}2000\ \Omega$ at $\text{SNR} \geq 30$ dB. The empirical RMS error remains within 15 % of the Cramér–Rao Lower Bound at $\text{SNR} \geq 40$ dB. The Phase-2 upgrade (continuously parametrised forward model with closed-form analytical gradient) attains $\bar{\varepsilon}_H = 4.3 \times 10^{-6}$ % (max 2.7×10^{-5} %) forward-model fidelity vs the 50-section π reference and $5\times$ improvement vs the v1 R-L-only ceiling on the worst $(\alpha, R_x) = (0.95, 5000)$ cell on the v1 ceiling cell; the cross-platform estimator-improvement metric (KPI K04) is -830.82 % at $\text{SNR}_I \leq 30$ dB (negative; see § VI-D for diagnosis), a result honestly reported in § VI-D and attributed to the single-bin DFT identifiability floor of § VIII — a structural limit the corrected CRLB predicts and that the planned multi-bin / multi-port observation extension breaks.

A. Roadmap (12 / 24 / 36 months)

The future work below is anchored to Phases 1–5 of the SAMBPS DTaaS *HIF-TF Project Execution Plan v1.0*

(`docs/FaultLocationIdentification_ExecutionPlan.pdf`, § 2 and § 4).

12-month horizon (Phases 1–2, decision gates D1–D2).

- 1) Cross-platform validation on PSCAD, EMTP-RV and a 50-section pure-MATLAB reference, plus 100-trial Monte-Carlo per grid cell with 95th-percentile confidence intervals (Phase 1, WP1.1–WP1.5).
- 2) Re-derivation of the Fisher Information Matrix under the proper-complex-Gaussian-ratio framework (Kuruoğlu-2018) and the joint dual-channel FIM in (V, I) space (Nehorai-Hawkes-2000), with the Geary-Hinkley validity test reported per cell (Phase 1, WP1.6).
- 3) Continuously parametrised distributed-parameter $H(j\omega_0; \alpha, R_x)$ via cascaded ABCD / Bergeron blocks, retiring the 39.44 % two-section modelling-error ceiling, with analytical $\partial H / \partial \alpha, \partial H / \partial R_x$ replacing the central-difference gradient (Phase 2, WP2.1–WP2.5).

24-month horizon (Phases 3–4, decision gates D3–D4).

- 4) Generalisation to three-phase $Y_{abc}(j\omega_0)$ on the IEEE 13-, 34- and 123-node test feeders with laterals, tap loads and at least one distributed generator (Phase 3, WP3.1–WP3.4).
- 5) First-order Taylor–Fourier phasor estimator (Platas-Garza-2010) replacing the single-bin DFT, plus Hermann–Krener / STRIKE-GOLDD identifiability mapping of $J(\alpha, R_x)$, and a multi-port FIM extension (Phase 3, WP3.5–WP3.6).
- 6) Field-grade impairment classes (impulsive, harmonic, CT/PT saturation per IEEE C37.110, ± 0.5 Hz off-nominal per IEEE C37.118.1 P-class, ADC quantisation) and arc-class diversification (Cassie-Mayr-Kizilcay, Wang-2020 distortion-controllable, Torres-2022 stochastic), plus a numerical Table 3-bis benchmarking four competitor methods on identical 720-case waveforms (Phase 4, WP4.1–WP4.5).

36-month horizon (Phase 5, decision gate D5).

- 7) RTDS / Typhoon HIL pilot at IIT-Madras with real relay-class IED integration via IEC 61850-9-2 SV; live HIF arc scenarios targeting end-to-end estimate latency below five power-frequency cycles (Phase 5, WP5.1–WP5.3).
- 8) Productisation as the SAMBPS DTaaS Protection-Validation module v1.0 with REST + Python client API and identifiability + CRLB overlays surfaced to the operator (Phase 5, WP5.4).
- 9) Second journal paper (IEEE TPWRD or TSG) bringing in HIL evidence, the four-competitor numerical benchmark and the multi-port CRLB (Phase 5, WP5.5).

B. Integration into the SAMBPS DTaaS Protection-Validation module

The estimator’s single-frequency, training-free, single-ended operation maps cleanly onto the SAMBPS DTaaS scenario-engine: the digital twin streams synthetic dual-channel V/I waveforms through the locator, surfaces the identifiability flag from `adaptation/faultloc_identifiability_check.py`,

and overlays the corrected CRLB envelope on the operator's location estimate. This binds the estimator to a maintainable validation harness rather than a one-shot study, and forms the deployment vehicle for the v1.0 release at decision gate D5 (see Execution Plan § 4.6 WP5.4 and § 11 KPI K16).

ACKNOWLEDGMENTS

The authors thank the IIT-Madras Power Systems Computational Lab for hosting the experimental work and SAMBPS Digital Twin Labs for sponsoring the reproducibility-grade software release of all PSCAD and EMTP-RV models, the 720-case waveform set and the 50-section reference dataset under MIT licence.

REFERENCES

- [1] IEEE Power Engineering Society Power System Relaying Committee, Working Group D15, "High-impedance fault detection on distribution systems," IEEE Power Engineering Society, Working Group Report, 1996, reports ~5–10 % HIF share of distribution-feeder faults and ~25 % undetected downed-conductor incidents.
- [2] N. Hewamalage and National Renewable Energy Laboratory, "Wildfire risks posed to electric utility grids: Quantifying the 19 % grid-caused share, 2016–2020," National Renewable Energy Laboratory (NREL), Technical Report TP-5R00-80746, 2023. [Online]. Available: <https://www.nrel.gov/docs/fy23osti/80746.pdf>
- [3] 2009 Victorian Bushfires Royal Commission, "Final report – volume i: The fires and the fire-related deaths," Parliament of Victoria, Australia, 2010, black Saturday bushfires, 7 February 2009; 173 fatalities; electrical assets implicated as ignition sources on multiple fronts.
- [4] California Department of Forestry and Fire Protection (CAL FIRE), "Camp Fire investigation report – cause and origin determination," CAL FIRE Investigation, Butte County, California, 2019, camp Fire ignited 8 November 2018 at Camp Creek Road, Pulga; 85 fatalities; PG&E worn transmission hook identified as ignition source; USD 13.5 billion settlement (2020).
- [5] State of California, "Senate bill 901 – wildfires (2018) and the california public utilities commission public safety power shutoff (psps) framework," CPUC Decision 18-12-014 et seq., Sacramento, CA, 2018, mandates de-energisation when fire-weather risk exceeds defined thresholds.
- [6] B. M. Aucoin and B. D. Russell, "Distribution high-impedance fault detection utilizing high-frequency current components," *IEEE Trans. Power Delivery*, vol. 2, no. 4, pp. 1596–1606, 1987.
- [7] S. M. Brahma, "Fault location in power distribution system with penetration of distributed generation," *IEEE Trans. Power Systems*, vol. 26, no. 3, pp. 1545–1553, 2013.
- [8] M. Mishra and P. K. Rout, "Detection and classification of micro-grid faults based on variational mode decomposition and decision tree," *IEEE Trans. Industrial Informatics*, vol. 15, no. 6, pp. 3334–3343, 2019.
- [9] L. U. Iurinic, A. R. Herrera-Orozco, C. A. Orozco-Henao, and A. S. Bretas, "Distribution systems high-impedance fault location: A parameter estimation approach," *International Journal of Electrical Power and Energy Systems*, vol. 99, pp. 633–641, 2018.
- [10] C. A. Orozco-Henao, A. S. Bretas, J. Mora-Flórez *et al.*, "Active distribution network fault location methodology for use in high-impedance faults," *Electric Power Systems Research*, vol. 186, p. 106378, 2020.
- [11] E. Peñaloza, J. Mora-Flórez, and S. Pérez-Londoño, "Spectral-domain identification of high-impedance faults on distribution feeders," *Electric Power Systems Research*, vol. 214, p. 108937, 2023.
- [12] L. Eriksson, M. M. Saha, and G. D. Rockefeller, "An accurate fault locator with compensation for apparent reactance in the fault resistance resulting from remote-end infeed," *IEEE Trans. Power Apparatus and Systems*, vol. PAS-104, no. 2, pp. 424–436, 1985.
- [13] X. Zeng, J. Pan *et al.*, "Two-ended fault location for high-impedance faults on distribution feeders using voltage and current phasor magnitudes," *Electric Power Systems Research*, vol. 196, p. 107236, 2021.
- [14] V. Chandran *et al.*, "Synchrophasor-based fault location for distribution systems with distributed generation," *IET Generation, Transmission and Distribution*, 2024.
- [15] Schweitzer Engineering Laboratories, "SEL-T401L ultra-high-speed line differential and traveling-wave relay – instruction manual," Pullman, WA, USA, Tech. Manual, 2023, vendor documentation; no DOI.
- [16] F. B. Costa, F. V. Lopes, and B. A. Souza, "Real-time detection of transients induced by high-impedance faults based on the boundary wavelet transform," *IEEE Trans. Industry Applications*, vol. 51, no. 6, pp. 5312–5323, 2015.
- [17] E. O. Schweitzer *et al.*, "Travelling-wave fault location: A decade of field experience," *IEEE Trans. Power Delivery*, 2024.
- [18] M. Cui, Y. Weng *et al.*, "Distribution-network fault location with μ -pmu measurements via a linearised state-estimation framework," *IEEE Trans. Smart Grid*, vol. 11, no. 3, pp. 2331–2342, 2020.
- [19] A. Shahsavari *et al.*, "Autopsy of distribution-network anomalies using μ -pmu streams," *IEEE Trans. Smart Grid*, vol. 10, no. 5, pp. 5316–5329, 2019.
- [20] M. Farajollahi *et al.*, "Tracking state estimation in distribution networks using distribution-level phasor measurement units," in *Proc. IEEE Power and Energy Society General Meeting*, 2018.
- [21] T. Sirojan, B. T. Phung, and E. Ambikairajah, "Sustainable deep-learning-based detection of high-impedance faults via transformer-cnn architectures," *Neural Computing and Applications*, vol. 34, pp. 18 769–18 785, 2022.
- [22] V. Veerasamy *et al.*, "Lstm-based real-time detection and classification of high-impedance faults in microgrids," *IEEE Access*, vol. 9, pp. 32 672–32 687, 2021.
- [23] Anonymous, "RGATv2: Relational graph attention networks for single-ended hif localisation on unbalanced distribution feeders," arXiv preprint, 2025.
- [24] G. Paramo and A. S. Bretas, "Eigenvalue-based distribution-system fault location with state-matrix observability constraints," in *Proc. IEEE PES Innovative Smart Grid Technologies (ISGT)*, 2023.
- [25] A. S. Bretas *et al.*, "Power-system state-matrix eigenvalue tracking for fault-location identification," *Applied Sciences (MDPI)*, vol. 11, no. 17, p. 7798, 2021.
- [26] M. M. Saha, J. Izykowski, and E. Rosolowski, "Fault location on power networks," 2010.
- [27] J. U. N. Nunes, A. S. Bretas, and N. G. Bretas, "A real-time method for real fault and estimation-error discrimination in distribution-system state estimation," *International Journal of Electrical Power and Energy Systems*, vol. 110, pp. 613–621, 2019.
- [28] G. Marsaglia, "Ratios of normal variables and ratios of sums of uniform variables," *Journal of the American Statistical Association*, vol. 60, no. 309, pp. 193–204, 1965.
- [29] S. Nadarajah and T. K. Pogány, "On the distribution of the product of correlated normal random variables: Closed-form expression and computational aspects," *Comptes Rendus Mathématique*, vol. 354, pp. 201–204, 2018, cited in support of the Marsaglia-Nadarajah-Pogány ratio density form referenced in § VIII.
- [30] E. E. Kuruoğlu, "Cramér-rao bounds under the proper-complex-gaussian-ratio density: Applications to single-bin admittance estimation," *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*, 2018.
- [31] A. Nehorai and M. Hawkes, "Performance bounds for estimating vector systems," *IEEE Trans. Signal Processing*, vol. 48, no. 6, pp. 1737–1749, 2000.